FinalProject - Mathew Scaria

2026-05-12

Introduction

Research Question: How do Vacancy Rates Affect Fair Market Rent?

Motivation: This is an important research topic because vacancy rates and fair market rent are closely tied to housing affordability, urban development, and economic stability. Understanding this relationship can help policymakers, landlords, and renters better understand how housing markets function.

Variables:

Dependent Variable: fmr_2bed (Rent for a 2 bedroom rental unit)

Independent Variable: vacancy_rate (The percent of rental units that are vacant for rent)

Controls:

total_housing_units (Total number of rental units this includes apartments, houses, etc)

median_income (People's Income in dollars)

households_with_children (The number of families that have children under 18)

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1. Importing my Data

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(stargazer)
```

```
##
## Please cite as:
```

```
##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
```

```
##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
FinalData <- read.csv("FinalProjectdataset.csv") #giving my data a name

head(FinalData) #This lets me see the first couple rows of data
```

```
##   fips_code    county_name state fmr_2bed vacancy_rate total_housing_units
## 1      1001 Autauga County    AL      977     8.928066               24731
## 2      1003 Baldwin County    AL     1206    26.358954              128518
## 3      1005 Barbour County    AL      740    22.406426               11702
## 4      1007    Bibb County    AL     1075    16.710671                9090
## 5      1009  Blount County    AL     1075    11.358045               24793
## 6      1011 Bullock County    AL      820    24.176548                4554
##   median_income households_with_children
## 1         69841                     5155
## 2         75019                    16692
## 3         44290                      962
## 4         51215                     1187
## 5         61096                     4405
## 6         36723                      402
```

```r
# Summary statistics for the variables I will use in my regressions
summary(select(FinalData, c('fmr_2bed', 'vacancy_rate', 'households_with_children', 'median_income', 't
```

```
##     fmr_2bed       vacancy_rate    households_with_children median_income
##  Min.   : 444.0   Min.   : 1.707   Min.   :     0.0         Min.   : 16170
##  1st Qu.: 795.0   1st Qu.: 9.514   1st Qu.:   652.5         1st Qu.: 54097
##  Median : 858.0   Median :15.011   Median :  1583.0         Median : 63128
##  Mean   : 965.3   Mean   :17.052   Mean   :  7157.6         Mean   : 64967
##  3rd Qu.:1024.5   3rd Qu.:21.603   3rd Qu.:  4475.0         3rd Qu.: 73083
##  Max.   :3293.0   Max.   :77.316   Max.   :594569.0         Max.   :178707
##  total_housing_units
##  Min.   :     47
##  1st Qu.:   5340
##  Median :  12375
##  Mean   :  44339
##  3rd Qu.:  30857
##  Max.   :3624084
```

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2. OLS Regression

This is my baseline OLS regression using all four variables on the full county-level dataset. I regress 2-bedroom Fair Market Rent on vacancy rate, total housing units, and median income.

```r
# OLS regression on the full dataset
reg_full <- lm(fmr_2bed ~ vacancy_rate + total_housing_units + median_income + households_with_children
summary(reg_full)
```

```
##
## Call:
## lm(formula = fmr_2bed ~ vacancy_rate + total_housing_units +
##     median_income + households_with_children, data = FinalData)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1404.17  -104.42   -26.70    69.14  1803.41
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               1.935e+02  1.611e+01  12.009  < 2e-16 ***
## vacancy_rate              1.323e+00  3.427e-01   3.862 0.000115 ***
## total_housing_units       1.218e-03  1.182e-04  10.310  < 2e-16 ***
## median_income             1.106e-02  2.046e-04  54.069  < 2e-16 ***
## households_with_children -3.302e-03  7.066e-04  -4.672  3.1e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 186.8 on 3206 degrees of freedom
## Multiple R-squared:  0.6122, Adjusted R-squared:  0.6117
## F-statistic:  1265 on 4 and 3206 DF,  p-value: < 2.2e-16
```

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3. Filtered Regression (High Income Counties)

Here I filter the dataset to only include counties where median household income is above $90,000. I want to see whether the relationship between vacancy rates and rent looks different in wealthier counties compared to the full sample.

```r
# Filter to high income counties only (median income above $90,000)
FinalData_high_income <- filter(FinalData, median_income > 90000)

cat("Number of counties in filtered dataset:", nrow(FinalData_high_income), "\n")
```

```
## Number of counties in filtered dataset: 248
```

```r
# OLS regression on the filtered high income dataset
reg_filtered <- lm(fmr_2bed ~ vacancy_rate + total_housing_units + median_income + households_with_chil
summary(reg_filtered)
```

```
##
## Call:
## lm(formula = fmr_2bed ~ vacancy_rate + total_housing_units +
##     median_income + households_with_children, data = FinalData_high_income)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -807.45 -255.50  -34.84  162.45 1676.58
##
## Coefficients:
```

```
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               8.067e+01  1.778e+02   0.454 0.650454
## vacancy_rate              4.872e+00  2.192e+00   2.223 0.027147 *
## total_housing_units       2.735e-03  4.463e-04   6.128 3.57e-09 ***
## median_income             1.234e-02  1.644e-03   7.508 1.14e-12 ***
## households_with_children -8.938e-03  2.308e-03  -3.873 0.000138 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 372 on 243 degrees of freedom
## Multiple R-squared:  0.3752, Adjusted R-squared:  0.3649
## F-statistic: 36.47 on 4 and 243 DF,  p-value: < 2.2e-16
```

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4. Preferred Regression (Log-Transformed Income)

My preferred regression uses the full dataset but log-transforms median income and rent. Income tends to have a non-linear relationship with rent — an extra dollar of income matters more at lower income levels than at higher ones. Log-transforming income captures this and typically improves model fit.

```r
# I logged income and fmr_2bed
FinalData <- mutate(FinalData, log_income = log(median_income), log_fmr = log(fmr_2bed))

# Preferred regression with log-transformed income
reg_preferred <- lm(log_fmr ~ vacancy_rate + total_housing_units + log_income + households_with_childre
summary(reg_preferred)
```

```
##
## Call:
## lm(formula = log_fmr ~ vacancy_rate + total_housing_units + log_income +
##     households_with_children, data = FinalData)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.26972 -0.10525 -0.02761  0.07927  0.89876
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.147e-01  1.178e-01   1.823   0.0684 .
## vacancy_rate              5.390e-04  2.917e-04   1.847   0.0648 .
## total_housing_units       7.502e-07  9.917e-08   7.565 5.04e-14 ***
## log_income                5.967e-01  1.055e-02  56.543  < 2e-16 ***
## households_with_children -1.266e-06  5.891e-07  -2.148   0.0318 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1589 on 3206 degrees of freedom
## Multiple R-squared:  0.6161, Adjusted R-squared:  0.6156
## F-statistic:  1286 on 4 and 3206 DF,  p-value: < 2.2e-16
```

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5. Stargazer Table

The table below displays all three regressions side by side for easy comparison.

```
#This table lets me create a singular table that lets me see all three of my regressions
#I used claude for this section to put the stargazer table together
stargazer(reg_full, reg_filtered, reg_preferred,
          type             = "latex",
          title            = "OLS Regression Results: Fair Market Rent (2-Bedroom)",
          dep.var.caption  = "",
          dep.var.labels = "2-Bedroom Fair Market Rent (USD)",
          covariate.labels = c("Vacancy Rate (Percent)",
                               "total housing units",
                               "Median Income (USD)",
                               "Log Median Income"),
          column.labels    = c("Full Sample", "High Income Counties", "Log Income and log Rent"),
          omit.stat        = c("f", "ser"),
          digits           = 3,
          notes            = "Data: HUD Fair Market Rent API and U.S. Census Bureau ACS 2023.",
          notes.align      = "l")
```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Wed, May 13, 2026 - 9:56:59 PM

```
library(tinytex)
```

```
## Warning: package 'tinytex' was built under R version 4.5.3
```

Table 1: OLS Regression Results: Fair Market Rent (2-Bedroom)

	2-Bedroom Fair Market Rent (USD)		log_fmr
	Full Sample	High Income Counties	Log Income and log Rent
	(1)	(2)	(3)
Vacancy Rate (Percent)	1.323***	4.872**	0.001*
	(0.343)	(2.192)	(0.0003)
total housing units	0.001***	0.003***	0.00000***
	(0.0001)	(0.0004)	(0.00000)
Median Income (USD)	0.011***	0.012***	
	(0.0002)	(0.002)	
Log Median Income			0.597***
			(0.011)
households_with_children	−0.003***	−0.009***	−0.00000**
	(0.001)	(0.002)	(0.00000)
Constant	193.520***	80.672	0.215*
	(16.115)	(177.812)	(0.118)
Observations	3,211	248	3,211
R^2	0.612	0.375	0.616
Adjusted R^2	0.612	0.365	0.616
Note:	*p<0.1; **p<0.05; ***p<0.01		
	Data: HUD Fair Market Rent API and U.S. Census Bureau ACS 2023.		